Referee Report on the Manuscript JASA-T&M-2025-0507: "Fast and Flexible Inference for Spatial Extremes"

The paper under view studies statistical max-stable and r-Pareto models for spatial extremes that allow for efficient inference via the so-called spectral likelihood. As already stated in the literature, the applicability of this method relies on the fact that the exponent measure of any vector $(Z(s_1), \ldots, Z(s_D))$ has all its mass in the interior of $[0, \infty)^D$. Thus, the authors provide an equivalent criterion for this, based on conditional distributions of the spectral process. Furthermore, they propose two new classes of models extending/adapting the well-known Brown–Resnick and extremal-t processes to this setting leading to the skewed Brown–Resnick and truncated extremal-t models, respectively. The capability of the new method and models are demonstrated in numerical experiments. Finally, they are used in two real data applications on extreme precipitation and extreme sea surface temperature.

The paper addresses an important and methodologically challenging question – efficient inference for high-dimensional extremes. It contains a number of novel contributions. My feeling, however, is that these novel elements are too limited to target a journal like *JASA*. More precisely:

- Even though the equivalent criterion in terms of conditional probabilities is new, the fact that the exponent measure (and angular measure) does not put any mass on the boundary if the spectral process is strictly positive seems to be direct consequence from other results in the literature, e.g., in the cited work by Dombry and co-authors such as Dombry and Ribatet (2015).

- The skewed Brown–Resnick process model is a straightforward extension of the multivariate Hüsler–Reiss model by Padoan (2011) and Beranger et al. (2019). Furthermore, to become a truly spatial model, the scale matrix Σ should be replaced by a (positive definite) scale function already in its definition.

- The definition of the truncation looks a bit strange. I guess that you mean

$$(\tilde{Y}(s_1), \ldots, \tilde{Y}(S_D)) \stackrel{d}{=} (Y(s_1), \ldots, Y(S_D)) \mid \min_{i=1}^{D} Y(s) > 0.$$

  However, this is not a truly spatial model in the sense that it does not directly extend to a continuous spatial domain S. In this case, an appropriate condition would be that $\inf_{s \in S} Y(s) > 0$;; however, this does not appear to lead to tractable finite-dimensional likelihoods.

Considering these reservations regarding the manuscript's primary methodological contributions, I believe that the results, as well as the model once it is fully spatial, merit publication; however, JASA T&M may not be the most suitable outlet.